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**EC-227-CBCS**

**B.E. IV Semester**

Examination, June 2020

**Choice Based Credit System (CBCS)**

**Communication Systems**

**Time : Three Hours**

**Maximum Marks : 60**

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume any missing data.

1. a) Find the Fourier transform of a rectangular pulse 2 seconds long with a magnitude of 10 volts.  
b) State and prove linearity and frequency convolution property of Fourier transform.
2. a) Determine the signal energy and signal power for  
i)  $f(t) = e^{-3|t|}$  and  
ii)  $f(t) = e^{-3t}$   
b) With the help of neat sketches, explain the generation and detection of SSB-SC signal.
3. a) An amplitude modulated signal is represented by  
 $\phi_{AM}(t) = 10 \cos(2\pi \cdot 10^6 t) + 5 \cos(2\pi \cdot 10^6 t) \cos(2\pi \cdot 10^3 t) + 2 \cos(2\pi \cdot 10^6 t) \cos(4\pi \cdot 10^3 t)$  volts.  
Find the various frequency components present, corresponding modulation indices and the bandwidth.  
b) Explain the necessity of modulation.

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4.
  - a) Draw and explain Super heterodyne Receiver.
  - b) Calculate the figure of merit in DSB-SC system.
5.
  - a) With the help of block diagram, describe the functioning of a broadcast FM receiver clearly bringing out the functional aspects of each one of the building blocks.
  - b) What is an Image frequency? How is image frequency rejection achieved?
6.
  - a) Draw the block diagram of producing FM from PM. Also explain and write its advantages.
  - b) Compare direct and indirect method of generation.
7.
  - a) Write down the classification of noise.
  - b) Explain the effect of noise on AM and FM receivers.
8. Write short note on any two:
  - a) Comparison between AGC and AFC
  - b) Discriminator method of FM detection
  - c) Synchronous generation and detection

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